MIL-S-001222G(SHIPS)
AMENDMENT-2
23 July 1979
SUPERSEDING
AMENDMENT-1
14 February 1977

MILITARY SPECIFICATION

STUDS, BOLTS, HEX CAP SCREWS, AND NUTS

This amendment forms a part of Military Specification MIL-S-001222G(SHIPS), dated 13 January 1976.

PAGE 2

* 2.1, under "SPECIFICATIONS, FEDERAL": Delete all reference to QQ-P-416 and QQ-Z-325 in their entirety.

* 2.1, under "SPECIFICATIONS, MILITARY", add:

"MIL-C-81751 - Coating, Metallic-Ceramic.
MIL-C-83488 - Coating, Aluminum, ION Vapor Deposited.
MIL-C-87115 - Coating, Immersion Zinc Flake/Chromate Dispersion."

* 2.2, under "AMERICAN SOCIETY FOR TESTING AND MATERIALS": Delete all reference to ASTM A 153, and add:

"ASTM B 454 - Mechanically Deposited Coatings of Cadmium and Zinc on Ferrous Metals.
ASTM B 633 - Electrodeposited Coatings of Zinc on Iron and Steel."

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2.2, under "AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)", add:

"B1.12 - Class 5 Interference-fit Thread."

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Table I: Footnote 3, delete second sentence.

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Table II: For Grade 5 steels over 1-1/2-inch diameter, Machined specimens, yield strength, delete "90" and substitute "58".

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Table III: For Grade 5 nuts 1/4 to 1 inch, types II and IV, delete "170" and substitute "72".

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* 3.3 through 3.3.2: Delete and substitute:

"3.3 Protective coatings.

"3.3.1 Carbon and alloy steels. When specified (see 6.1 and 6.2.1), carbon and alloy steel fasteners shall be coated as follows:

Metallic-ceramic coating: MIL-C-81751, type I, class 4
Ion vapor deposited aluminum: MIL-C-83488, type II, class 3
Zinc flake/chromate dispersion: MIL-C-87115, class 3
Electrodeposited zinc: ASTM B 633, type II, class Fe/Zn 13
Mechanically deposited zinc: ASTM B 454, class 13

"3.3.1.1 When zinc coating is specified, the coating may be either electrodeposited or mechanically deposited."

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3.4.1: Add "Class 5 interference-fit threads shall be in accordance with ANSI B1.12."

* 3.4.3.1: Delete.

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3.5.1: Delete and substitute:

* "3.5.1 Material identification. Fasteners 1/4 inch size and over (except for grades 1 and 2) shall be marked to identify the material. In addition, all externally threaded grade 500 (nickel-copper-aluminum) and grade T7 (titanium) fasteners 1/4 inch and larger shall be marked with the heat or lot number and the manufacturer's symbol. Marking shall be of permanent type and shall be raised or depressed. Stamping shall be made with low stress (round radius) die stamps. Marking shall be applied to the head of the hex cap screws, the nut end of tap end studs, one end of continuous studs, and the top face (not washer face) of nuts. Manufacturer's identification symbols shall be applied to fasteners procured to lot definition "A". Manufacturer's identification symbols may be omitted for fasteners procured to lot definition "B" (see 4.2)."

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3.5.1.1: Delete.

* 4.1: Delete and substitute:

"4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements."

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Table VIII: For Mechanical tests, delete "MIL-STD-105, special inspection level S-1, AQL = 1.5" and substitute "2/".

After footnote 1/, add:

*2/

Lot size (pieces)	Sample size	Acceptable defects
50 and less	2	0
51 - 500	3	0
501 - 3500	5	0
Over 3500	8	0

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* 6.1.1: Delete and substitute:

"6.1.1 Protective coatings. The characteristics of the various coatings are contained in the applicable plating specifications. Recommended applications are summarized in table XIa."

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Add table XIa.

"TABLE XIa. Coatings for various applications.

Coating type	Base material	Upper temperature limit 1/	Required salt spray test (min.)	Environment
Metallic-ceramic MIL-C-81751, type 1, class 4.	Carbon and alloy steels 3/8 inch and larger	1000°F	1000 hrs	High temperature and severe corrosion
IVD Aluminum MIL-C-83488, type II, class 3	Carbon and alloy steels. No size limitation.	925°F	336 hrs	Moderate temperature and corrosion
Immersion zinc/flake MIL-C-87115 class 3	Carbon and alloy steels. No size limitation.	650°F	336 hrs	Moderate temperature and corrosion
Zinc coating ASTM B 454, class 13 or ASTM B 633, type II, class Fe/Zn 13	Grades 2 and 5 only (Not recommended for high strength steels)	300°F	96 hrs	Ambient temperature and mild corrosion

1/ Maximum service temperature applies to coating alone and not to the base metal (see table XI)."

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6.2.1(c): Delete and substitute:

"(c) Type of coating required (see 3.3.1 and 6.1.1)."

NOTE: The margins of this amendment are marked with an asterisk (*) to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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